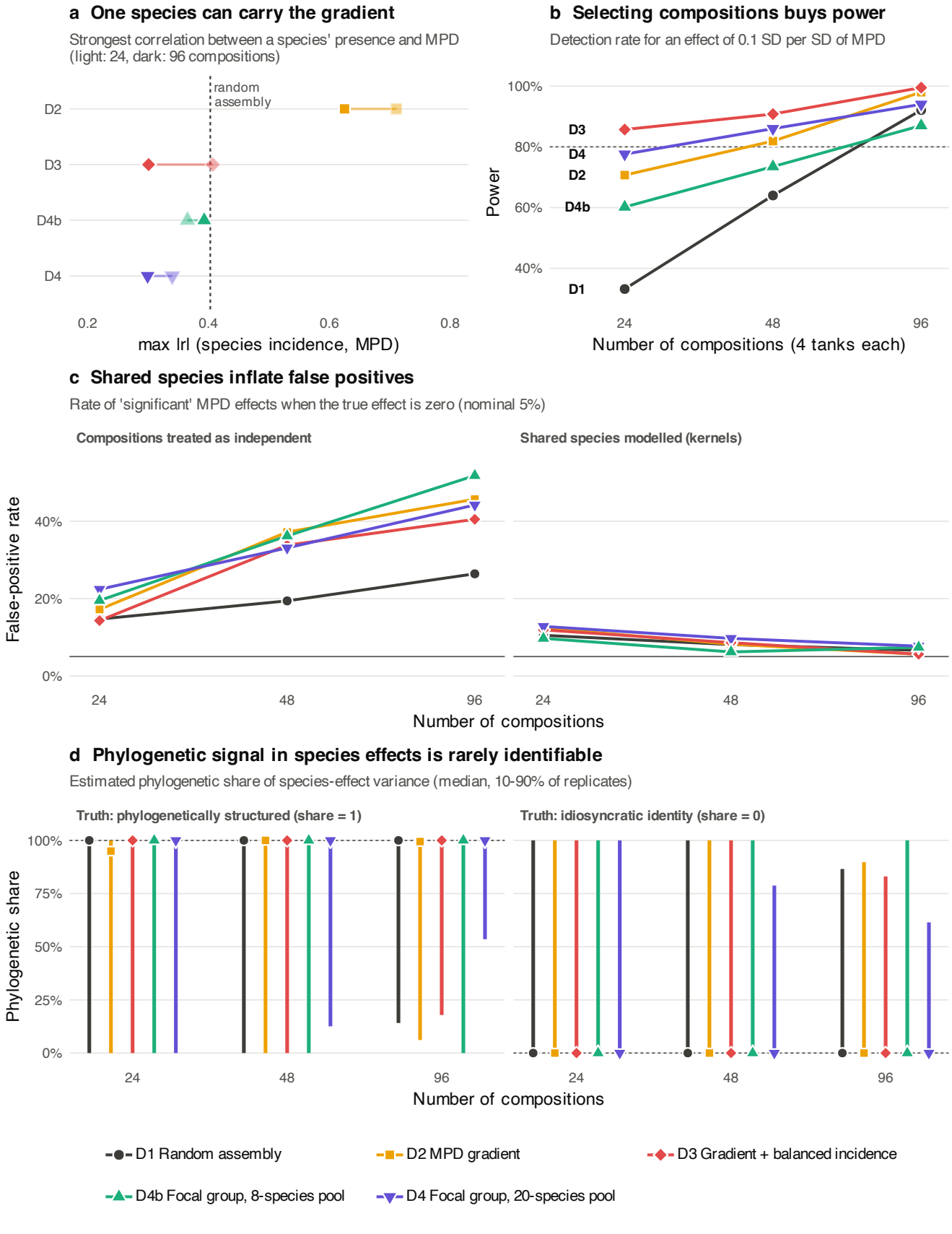
a One species can carry the gradient
Strongest correlation between a species' presence and MPD
(light: 24, dark: 96 compositions)
random
assembly
D2
D3
D4b
D4
0.2
0.4
0.6
0.8
max lrl (species incidence, MPD)
b Selecting compositions buys power
Detection rate for an effect of 0.1 SD per SD of MPD
100%
80%
60%
40%
Power
D3
D4
D2
D4b
D1
24
48
96
Number of compositions (4 tanks each)
c Shared species inflate false positives
Rate of 'significant' MPD effects when the true effect is zero (nominal 5%)
Compositions treated as independent
Shared species modelled (kernels)
False-positive rate
40%
20%
0%
24
48
96
Number of compositions
d Phylogenetic signal in species effects is rarely identifiable
Estimated phylogenetic share of species-effect variance (median, 10-90% of replicates)
Truth: phylogenetically structured (share = 1)
Truth: idiosyncratic identity (share = 0)
Phylogenetic share
100%
75%
50%
25%
0%
24
48
96
Number of compositions
D1 Random assembly
D2 MPD gradient
D3 Gradient + balanced incidence
D4b Focal group, 8-species pool
D4 Focal group, 20-species pool